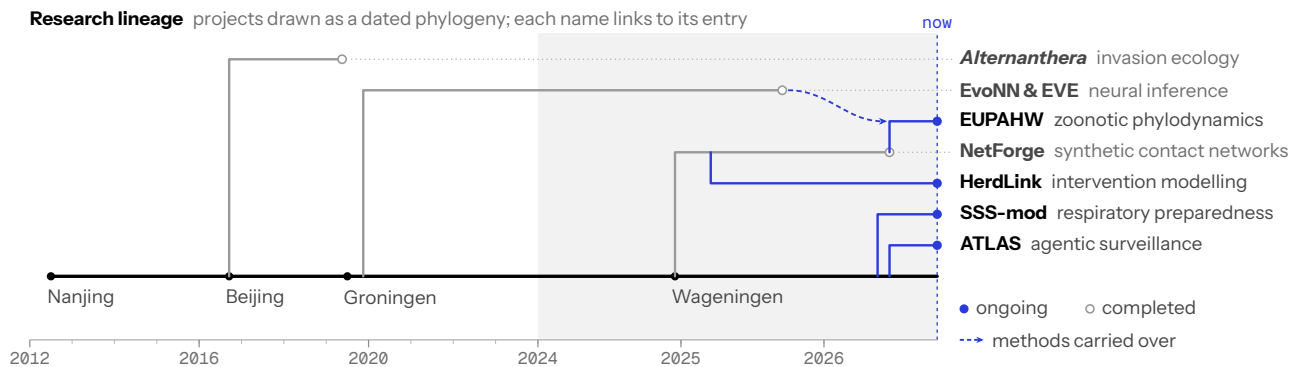


Tianjian Qin PhD

Computational biologist & AI/ML researcher
Postdoctoral researcher, Wageningen University & Research

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Life is interconnected. I build tools to understand how.



Profile

Computational biologist developing machine learning methods, stochastic models and research software. My work spans neural inference from phylogenies, livestock movement networks and infectious disease modelling, with reproducible computing and interactive tools for surveillance and simulation.

Focus

Computational biology, AI/ML, infectious disease modelling, network science, phylodynamics, One Health, research software.

Languages

Chinese, native. English, proficient. Dutch, A2 certified with strong reading. German, A2.

Research experience

2025 – now

Postdoctoral Researcher

Wageningen University & Research, Infectious Disease Epidemiology Group

Develop computational methods and research software for One Health and pandemic preparedness. Work across livestock movement networks, hepatitis E surveillance and respiratory transmission models; build synthetic populations, reproducible analysis pipelines and interactive tools for scenario comparison.

2019 – 2026

Doctoral Researcher in Theoretical & Computational Biology

University of Groningen, Groningen Institute for Evolutionary Life Sciences

Developed simulation and inference methods for evolutionary diversification; built research packages in R, C++ and Python with CPU/GPU computing, testing, documentation and open releases.

Education

2019 – 2026

PhD in Theoretical Biology

University of Groningen, Netherlands

Thesis: [Diversification Models and Neural Inference](#). Supervisors: Rampal Etienne, Luis Valente and Koen van Benthem.

2016 – 2019

MSc in Wetland Ecology

Beijing Forestry University, China

Supervisors: Hongli Li and Feihai Yu.

2012 – 2016

BSc in Marine Biology

Nanjing Normal University, China

Honours

2025

Denise Kirschner Best Student Paper Prize

Journal of Theoretical Biology, inaugural prize for graduate student research

2017–2019

First-Class Academic Performance Award, Beijing Forestry University

2018

Graduate Academic Innovation Award, Beijing Forestry University

Current projects

Jun 2026–now

ATLAS agentic tracking and longitudinal analysis for surveillance

Developer and maintainer, Wageningen University & Research

Develop a Python workflow combining LLM extraction, scientific review and source provenance for outbreak surveillance. Build versioned evidence, longitudinal comparisons and an interactive interface for geographic, reporting-network and One Health exploration.

Methods Python, agentic AI, evidence review

Software qtj.me/atlas (live), github.com/EvoLandEco/ATLAS

Jun 2026–now

EUPAHW zoonotic pathogens across pigs, wild boar and humans

Researcher, Wageningen University & Research

Develop phylodynamic and network analyses linking livestock movements, wildlife and landscape data to investigate transmission pathways of zoonotic pathogens, such as hepatitis E virus and swine influenza, and inform One Health surveillance.

Methods Phylodynamics, movement networks, One Health surveillance

Programme European Partnership on Animal Health and Welfare, SOA26

May 2026–now

SSS-mod preparedness package for future respiratory outbreaks

IMPULS preparedness application, Wageningen University & Research

Develop data preparation, recalibration, model output and interface workflows supporting reuse of a setting-specific respiratory transmission model.

Methods Python and FastAPI, React and TypeScript, population synthesis

Programme ZonMw, Implementatie-impuls COVID-19 programma

Mar 2025–now

HerdLink interactive intervention modelling

Author and developer, Wageningen University & Research

Developed within IMBIT and deployed as a public web platform linking Dutch livestock trade networks, geographic views and disease simulations. Implemented dated route and regional movement restrictions, seven scenario presets, and paired baseline and intervention comparisons of epidemic trajectories, regional burden and retained trade for scenario comparison and decision support.

Methods JavaScript, React, D3.js, transmission simulation

Software herdlink.nl (live), github.com/EvoLandEco/herdlink-web

Programme IMBIT, ZonMw Kennisprogramma Pandemische Paraatheid

2026

NextdAI integrated AI-driven early detection system

NWO KIC, Early Detection of (Emerging) Infectious Disease Outbreaks

Co-develop AI architecture and data infrastructure for heterogeneous surveillance, anomaly detection, neurosymbolic characterisation, forecasting and decision support.

Methods AI architecture, surveillance data integration, forecasting

proposal

Earlier research & software

Dec 2024–Jun 2026

NetForge synthetic livestock contact networks

Wageningen University & Research, IMBIT

Developed within the IMBIT pandemic preparedness programme, combining stochastic block models with contact history selection to generate synthetic livestock contact networks; benchmarked them against temporal reachability and simulated infection dynamics.

Methods Stochastic block models, temporal networks, epidemic simulation

Software qtj.me/NetForge, github.com/EvoLandEco/NetForge

Publications *bioRxiv* preprint (2026), doi:10.64898/2026.09.15.751711

Nov 2019 – Sep 2025

EvoNN & EVE neural inference for evolutionary biology

University of Groningen

Designed ensemble neural networks that learn from tree structure, branching times and summary statistics, alongside stochastic diversification models and high-performance simulators.

Methods Neural inference, stochastic simulation, R, C++ and Python

Software github.com/EvoLandEco/EvoNN, github.com/EvoLandEco/evesim

Publications [Systematic Biology \(2026\)](#), [Journal of Theoretical Biology \(2025\)](#)

Sep 2016 – May 2019

Understanding the invasion of *Alternanthera philoxeroides* in China

MSc research, Beijing Forestry University

Studied phylogenetic diversity, spatial distribution and wetland community resistance to invasion. Combined surveys across 18 Chinese provinces with controlled experiments, image analysis and phylogenetic workflows to examine community composition and environmental effects.

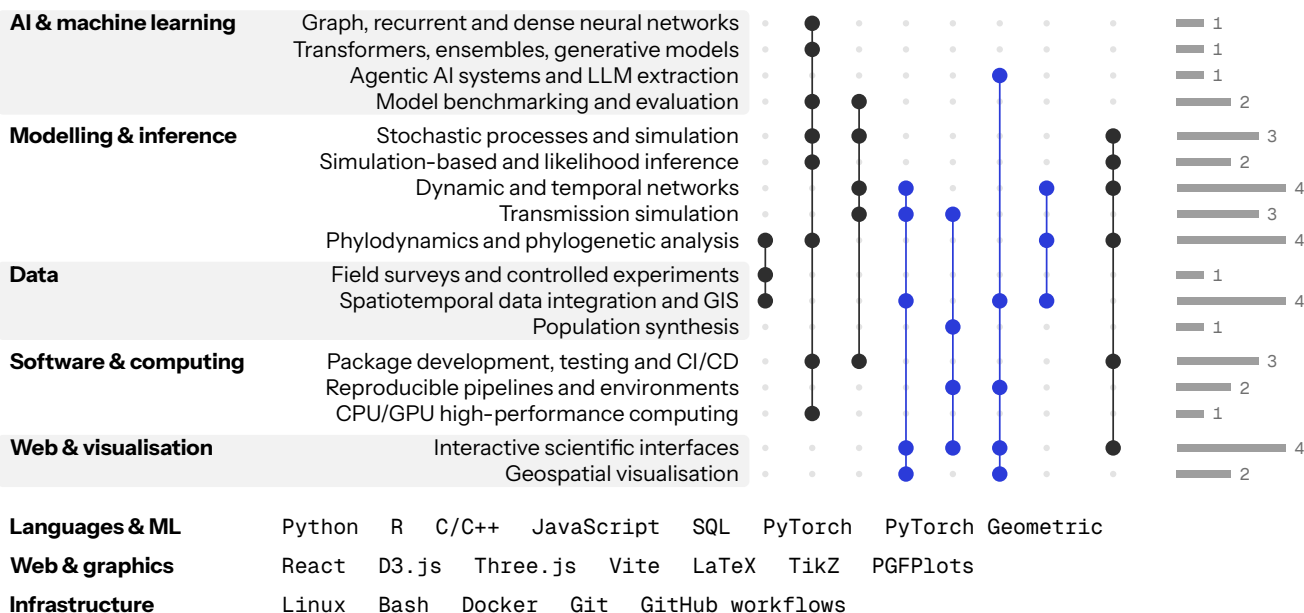
Methods Field surveys, controlled experiments, phylogenetic analysis

Publications [Plant Biology \(2020\)](#), [Marine and Freshwater Research \(2018\)](#)

Technical expertise

Methods by project

columns follow the research lineage; names and dots link to entries



Software contributions

Developer

[NetSpectra](#), [EvoLab](#), [miniape](#). Interactive network dynamics, lineage simulation, tree reconstruction and browser-based phylogenetic data handling.

Collaborator

[treestats](#), [DDD](#), [DAISIE](#). Phylogenetic statistics, data structures, simulation and inference for diversification and island biogeography.

Teaching, mentoring & service

Teaching

Seminars and workshops at Groningen, WUR and Wageningen Bioveterinary Research on island biogeography, stochastic modelling, phylogenetics, neural networks and network statistics.

Mentoring

BSc/MSc students and junior PhD researchers; pair programming on test-driven development and scientific software design.

Peer review

Systematic Biology, *PLOS Computational Biology*, *Epidemics*, *BMC Public Health* and *BMC Plant Biology*.

Leadership

Vice President, Graduate Student Council (2017–2018); student representative, Wetland Ecology Group (2016–2019), Beijing Forestry University.

Peer-reviewed articles

- 2026 Parameter estimation from phylogenetic trees using neural networks and ensemble learning.
Qin, T., van Benthem, K. J., Valente, L.[†] & Etienne, R. S.[†] *Systematic Biology* 75(2), 344–365
[doi:10.1093/sysbio/syaf060](https://doi.org/10.1093/sysbio/syaf060)
- 2025 Impact of evolutionary relatedness on species diversification and tree shape.
Qin, T., Valente, L.[†] & Etienne, R. S.[†] *Journal of Theoretical Biology* 598, 111992
[doi:10.1016/j.jtbi.2024.111992](https://doi.org/10.1016/j.jtbi.2024.111992)
- 2022 Relative abundance of invasive plants more effectively explains the response of wetland communities to different invasion degrees than phylogenetic evenness.
Sun, K., Liu, X.-S., **Qin, T.-J.**, Jiang, F., Cai, J.-F., Shen, Y.-L., A, S.-H. & Li, H.-L.
Journal of Plant Ecology 15(3), 625–638 [doi:10.1093/jpe/rtab074](https://doi.org/10.1093/jpe/rtab074)
- 2020 Phylogenetic diversity is a better predictor of wetland community resistance to *Alternanthera philoxeroides* invasion than species richness.
Qin, T., Zhou, J., Sun, Y., Müller-Schärer, H., Luo, F., Dong, B., Li, H. & Yu, F.-H.
Plant Biology 22(4), 591–599 [doi:10.1111/plb.13101](https://doi.org/10.1111/plb.13101)
- 2019 Spatial environmental heterogeneity may drive functional trait variation in *Hydrocotyle vulgaris* (Araliaceae), an invasive aquatic plant.
Wan, J.-Z., Wang, M.-Z., **Qin, T.-J.**, Bu, X.-Q., Li, H.-L. & Yu, F.-H. *Aquatic Biology* 28, 149–158
[doi:10.3354/ab00716](https://doi.org/10.3354/ab00716)
- 2019 Growth traits of the exotic plant *Hydrocotyle vulgaris* and the evenness of resident plant communities are mediated by community age, not species diversity.
Qin, T.-J.^{*}, Guan, Y.-T.^{*}, Quan, H., Dong, B.-C., Luo, F.-L., Zhang, M.-X., Li, H.-L. & Yu, F.-H.
Weed Research 59(5), 377–386 [doi:10.1111/wre.12373](https://doi.org/10.1111/wre.12373)
- 2018 Sediment type and nitrogen deposition affect the relationship between *Alternanthera philoxeroides* and experimental wetland plant communities.
Qin, T.-J., Guan, Y.-T., Zhang, M.-X., Li, H.-L. & Yu, F.-H.
Marine and Freshwater Research 69(5), 811–822 [doi:10.1071/MF17335](https://doi.org/10.1071/MF17335)
- 2018 Effects of water regime on the growth of the submerged macrophyte *Ceratophyllum demersum* at different densities.
Liu, L., Guan, Y.-T., **Qin, T.-J.**, Wang, Y.-Y., Li, H.-L. & Zhi, Y.-B.
Journal of Freshwater Ecology 33(1), 45–56 [doi:10.1080/02705060.2017.1422043](https://doi.org/10.1080/02705060.2017.1422043)

Preprints

- 2026 From movement to spread: generating livestock contact networks that preserve infection dynamics.
Qin, T., Atamer Balkan, B., Schmid, B. V. & ten Bosch, Q. A. *bioRxiv* preprint
[doi:10.64898/2026.09.15.751711](https://doi.org/10.64898/2026.09.15.751711)
- 2026 Identifying evolutionary relatedness effects on diversification from phylogenies using neural networks.
Qin, T., van Benthem, K., Valente, L.[†] & Etienne, R. S.[†] *bioRxiv* preprint
[doi:10.64898/2026.01.12.699140](https://doi.org/10.64898/2026.01.12.699140)

^{*} Equal contribution. [†] Joint senior authors.

Talks & workshops

- Oct 2026 **Trade networks, Transmission, Intervention**
Led workshop, WUR MIND Day 2026, Wageningen
- Aug 2026 **From livestock movements to infection dynamics: mechanism-informed learning for synthetic contact networks**
Oral presentation, [ModAH Conference 2026](#), Nantes, France. With B. Atamer Balkan, B. Schmid and Q. ten Bosch.